

BHGap: A Deep Iterative Prompting and Multi-stage Alignment Framework for Dynamic Facial Expression Recognition

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Abstract

Dynamic Facial Expression Recognition (DFER), as a crucial part of affective computing, has broad applications in many areas such as human-computer interaction and social media content analysis. Effectively integrating multimodal information, particularly audio-visual signals, remains the core challenge. However, existing approaches are generally constrained by two major challenges: (1) shallow and static fusion mechanisms, which fail to capture the dynamic co-evolution of audio-visual features during deep interaction; (2) implicit and coarse alignment strategies, which are insufficient to bridge the modality gap caused by heterogeneous feature distributions. To address these issues, we propose a novel framework, *BHGap*, which integrates deep iterative prompt generation with multi-stage feature alignment and fusion. The key idea is to reformulate audio-visual collaboration from a one-shot fusion event into a continuous, reciprocal generation process that spans every layer of frozen backbone encoders. Specifically, we design a State Space Model (SSM)-based cross-modal prompt generator that dynamically produces "guidance prompts" for the counterpart modality at each encoding layer, thereby enabling deep and fine-grained feature co-evolution. Beyond encoding, we further introduce a coarse-to-fine multi-stage alignment module: at the macro level, low-rank adversarial alignment is employed to establish spatio-temporal congruity between audio and video while reducing global distributional discrepancies; at the micro level, Maximum Mean Discrepancy (MMD) constraints combined with implicit differentiation optimization ensure fine-grained statistical consistency and semantic alignment. Extensive experiments on the public DFEW and MAFW datasets demonstrate that our method achieves *state-of-the-art* performance, offering a new paradigm of deep iterative fusion and explicit alignment for multimodal emotion recognition. Code is available at <https://github.com/NDYZD666/-public-BHGap>.

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CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; • **Information systems** → **Multimedia and multimodal retrieval**.

Keywords

Dynamic Facial Expression Recognition, Audio-Visual Emotion Recognition, Multimodal Learning, Modality Alignment

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1 Introduction

Dynamic Facial Expression Recognition (DFER) has evolved from static facial expression recognition to complex emotional states conveyed through human facial dynamics in video sequences. This task is of great significance across diverse applications, including mental health assessment [15], web-based social media content analysis [11], intelligent education [17], and human-computer interaction [26]. Early studies primarily leveraged the intuitively visual properties of the DFER task, exploring approaches that extended from "static" images to "dynamic" sequences, and achieved promising results [41]. More recently, a growing body of research has highlighted that emotional expression in real-world contexts is inherently multimodal [21], typically manifested not only through facial muscle movements but also through auditory cues such as prosody and intonation. As a result, audio-visual emotion recognition (AVER) has emerged as a mainstream paradigm, where the effective integration of audio and visual modalities has become a core driving force for achieving good DFER systems.

However, achieving effective audio-visual synergy remains challenging due to the inherent modality gap. Visual signals are spatially dense pixel representations, whereas auditory signals are temporally dense feature sequences, making naive fusion strategies inadequate. Simple concatenation or shallow fusion often fails to capture deep semantic correlations and may even degrade performance due to modality conflicts. Existing fusion methods are

typically categorized into early, late, and hybrid fusion [2]. Early fusion integrates low-level features but is susceptible to heterogeneity and noise amplification [28]. Late fusion operates at the decision level but neglects intermediate cross-modal interactions [1]. Hybrid fusion methods introduce interactions via concatenation or cross-attention [33], yet these "one-shot" static paradigms are insufficient for modeling continuous and fine-grained multimodal cooperation. Moreover, most existing approaches rely on implicit alignment driven solely by the final classification loss [25, 38], which often leads to suboptimal solutions. Furthermore, most existing models rely on implicit alignment driven by the final classification loss [25, 38]. This constraint often traps models in suboptimal solutions. In summary, three bottlenecks in current audio-visual recognition research are concluded: (1) **Shallow and static fusion mechanisms** — lacking deep, dynamic interactions throughout the feature extraction process. (2) **Implicit and coarse alignment strategies** — unable to explicitly bridge cross-modal feature distributions and guarantee fine-grained semantic consistency. (3) **Insufficient spatio-temporal synergy modeling** — ineffective in capturing joint temporal-spatial correspondences and co-evolution across modalities.

To address these challenges, we propose a novel framework *BHGap*, that integrates Deep Iterative Prompt Generation with Multi-Stage Feature Alignment. The central idea is to transform cross-modal synergy from a one-time fusion event into a continuous, mutually guided generation process spanning every layer of the audio and visual encoders. Moreover, prior to the final classification stage, we introduce a carefully designed coarse-to-fine alignment module that explicitly calibrates the feature distributions of both modalities. Specifically, to address the insufficiency of cross-modal interactions and the limitations of static fusion, for deep dynamic fusion, we design an innovative layer-wise iterative prompt generator embedded after each Transformer block of the frozen audio-visual encoders. Leveraging the strong sequence modeling capacity of State Space Model (SSM) [14], the generator produces guidance prompts tailored to each modality, informed by the complementary modality. This iterative process establishes a tightly coupled "dialogue" mechanism that enables mutual learning through recurrent feedback across layers. We propose a two-stage strategy that simultaneously tackles the problem of multimodal spatio-temporal cooperative modeling and achieves explicit fine-grained cross-modal alignment to eliminate modality gaps. In the first stage, a low-rank adversarial training mechanism enforces macro-level spatio-temporal congruity between video and audio modalities; in the second stage, implicit differentiation with Maximum Mean Discrepancy (MMD) [13] loss further narrows the feature gap at the statistical moment level, ensuring semantic consistency.

The main contributions of this paper are as follows:

- To overcome the limitations of shallow and insufficiency of cross-modal interactions, we propose an SSM-based deep iterative cross-modal co-evolution framework (SDIC) embedded throughout the feature extraction process. By dynamically generating cross-modal prompts at each encoder layer, our approach transforms the conventional shallow, static

fusion paradigm into a deep, dynamic, and fine-grained co-evolution paradigm.

- We design a multi-stage, coarse-to-fine alignment module (MSC2F) that innovatively combines adversarial alignment with implicit differentiation. By leveraging low-rank transformations and MMD loss, this module achieves dual alignment at both macro distribution and micro statistical levels, effectively bridging the modality gap.
- To the best of our knowledge, we are the first to introduce SSM into cross-modal guidance generation, providing an efficient and effective architecture for handling multimodal interactions.
- Extensive experiments conducted on the DFEW and MAFW datasets demonstrate that our model achieves state-of-the-art performance, validating the effectiveness of our proposed framework.

The structure of this paper is arranged as follows. Section 2 reviews related work on DFER. Section 3 details our model *BHGap*. Section 4 shows the experiments and the results. Section 5 is the conclusion. The technical details of SDIC and MSC2F, as well as the model hyperparameter experiments, are presented in the Appendix section.

2 Related Work

Multimodal Dynamic Facial Expression Recognition. Human emotional expressions are inherently multimodal, involving both facial movements and vocal cues. Consequently, recent DFER studies have shifted from unimodal video inputs to multimodal frameworks, such as audio-visual emotion recognition [40]. A central challenge is the effective integration of heterogeneous modalities, and existing methods are generally categorized into early fusion, late fusion, and hybrid fusion.

Early fusion combines multimodal features at the input level. For example, MV-LSTM [30] concatenates raw features, while TFN [39] models interactions via tensor fusion. Despite their simplicity, these methods often suffer from modality interference caused by structural and distributional discrepancies.

Late fusion independently processes each modality and integrates high-level representations. Representative approaches include T-ESFL [25], which concatenates features; AMH [38], which employs cross-modal attention; and CLIP-based models [29], such as DFER-CLIP [42], which aligns visual and textual features via contrastive learning. While alleviating low-level incompatibility, these methods typically overlook deeper semantic interactions.

Hybrid fusion introduces cross-modal interactions during feature extraction. Methods such as TMEP [40] adopt layer-wise cross-modal attention, and MMA-DFER [7] employs a bottleneck-based interaction mechanism. Although hybrid approaches better capture complementary information, most rely on static or coarse interactions, limiting their ability to model dynamic synergy and fine-grained correspondences, and may even suppress weaker modalities.

These limitations are closely related to the modality gap problem. In AVER tasks, visual features emphasize spatial structures, whereas auditory features focus on temporal and spectral patterns, leading

to substantial distributional and semantic discrepancies. Many studies rely on implicit alignment through classification loss, resulting in pseudo-alignment dominated by information-rich modalities. Inspired by CLIP, methods such as CLIPER [23], DK-CLIP [24], and DFER-CLIP [42] employ contrastive learning for semantic alignment, while HiCMAE [32] further introduces hierarchical alignment from low- to high-level features. Nevertheless, most approaches remain limited to global alignment and neglect fine-grained spatio-temporal correspondences, which are critical for distinguishing subtle expressions in DFER.

Temporal Modeling in Dynamic Facial Expression Recognition. DFER extends static facial expression recognition from single images to continuous video sequences, introducing the temporal dimension into emotion recognition and making temporal modeling a core research problem.

Early studies commonly employed recurrent neural networks (RNNs, LSTMs) to model temporal dependencies, but these approaches suffer from gradient vanishing and limited efficiency. The advent of the Transformer has largely mitigated these issues. Representative methods such as Former-DFER [41], DFER-CLIP [42], and MMA-DFER [7] leverage Transformer architectures for effective temporal modeling. Other works, including M3DFEL [34] and DK-CLIP [24], decompose frame sequences and apply local and global temporal attention, while FineCLIPER [4] introduces temporal adapters into frozen pretrained encoders. Additionally, self-supervised approaches such as MAE-DFER [31] employ temporal masking to learn spatio-temporal representations.

Nevertheless, many existing methods still model temporal and spatial features independently, limiting their ability to capture the continuity and co-evolution of emotions, which are inherently influenced by dynamic dependencies across adjacent time steps.

3 Methodology

3.1 Problem Formulation

Given a video segment $\mathcal{V}$ containing F ordered frames and its corresponding audio spectrogram $\mathcal{A}$, each annotated with an emotion label $y \in \{1, 2, \dots, K\}$, the goal of DFER is to learn a mapping function $\mathcal{F}_\theta$ such that:

$$\mathcal{F}_\theta : (\mathcal{A}, \mathcal{V}) \rightarrow y, \quad (1)$$

where θ denotes the model parameters, and K is the total number of emotion categories. This task requires joint representation learning over heterogeneous modalities under spatio-temporal constraints, aiming at accurate emotion recognition.

3.2 Model Overview

This section presents an overview of our *BHGap* model, a novel multimodal audio-visual emotion recognition framework tailored for the DFER task. As illustrated in Fig. 1, *BHGap* consists of three components: (1) **Frozen Audio-Visual Backbone Encoders** (Section 3.3), which independently extract modality-specific features for audio and video inputs; (2) **SSM-based Deep Iterative Cross-modal Co-evolution Framework (SDIC)** (Section 3.4), which facilitates dynamic, fine-grained co-evolution between modalities; (3) **Multi-Stage Coarse-to-Fine Feature Alignment (MSC2F)** (Section 3.5), designed to mitigate the heterogeneity gap during

fusion. The video and audio features are first fed into their respective frozen encoders for feature extraction. Through the SDIP module, interactive co-learning and domain transfer are performed at each encoder layer. The output features from the dual-branch encoders are then passed into the MSC2F module for cross-modal alignment, which further eliminates modality discrepancies. After alignment, the multimodal features are integrated within a stack of Transformer layers for deep feature fusion. Finally, the fused representation is fed into a classifier to produce the recognition results. Audio-Visual Backbone Encoders, SDIC, and MSC2F are expanded in Sections 3.3, 3.4, and 3.5, respectively. The loss functions of the proposed model will be detailed in Section 3.6.

3.3 Audio-Visual Backbone Encoders

Since the Vision Transformer (ViT) architecture has demonstrated strong generalization across both visual and auditory classification tasks, our framework adopts two independent, pre-trained ViT models as frozen backbones for video and audio modalities, respectively. The video input $\mathcal{V} \in \mathbb{R}^{B \times F \times L_V \times D_v}$ and the audio input $\mathcal{A} \in \mathbb{R}^{B \times L_A \times D_a}$ are embedded into modality-specific sequences, where B is the batch size, F is the frame count, L_V and L_A are the sequence lengths for video patches and audio spectrogram tokens, and D_v, D_a are the corresponding feature dimensions.

Considering that DFER datasets are typically limited in scale and exhibit noise, inspired by prior works such as S2D [5] and MMA-DFER [7], we select the approach of leveraging ViT [9] pre-trained on large-scale datasets from domains closely related to DFER, followed by transfer learning through fine-tuning on the DFER dataset. For simplicity, we employ the pre-trained weights from MAE-Face [27] and AudioMAE [18] to initialize the visual and audio encoders, respectively. The potential of other, more powerful pre-training models can be explored in subsequent work. Importantly, all parameters of the backbone encoders remain frozen during training.

This design ensures that *BHGap* leverages robust and transferable pre-trained features, allowing the subsequent modules to concentrate on fine-grained cross-modal interaction and feature alignment, rather than re-learning modality-specific representations from limited DFER data.

3.4 SSM-based Deep Iterative Co-evolution (SDIC)

Conventional audio-visual models often adopt static fusion strategies, such as concatenating features or applying cross-attention only at the top layers of the encoders. However, these models overlook the necessity of cross-modal information exchange during the deeper stages of feature extraction. In contrast, emotional perception is a dynamic process where visual and auditory cues continuously influence and co-evolve with each other over time.

To emulate this dynamic process, we design a fine-grained deep iterative co-evolution framework, whose core is a Cross-Modal Prompt Generator (CMPG) based on SSM [14], embedded between each pair of consecutive layers of the frozen backbone encoders. This framework transforms cross-modal fusion from a one-time operation into a continuous, mutually guided generative process across all encoder layers. Specifically, instead of directly passing the output of the ℓ -th encoder layer into the $(\ell + 1)$ -th layer, the feature

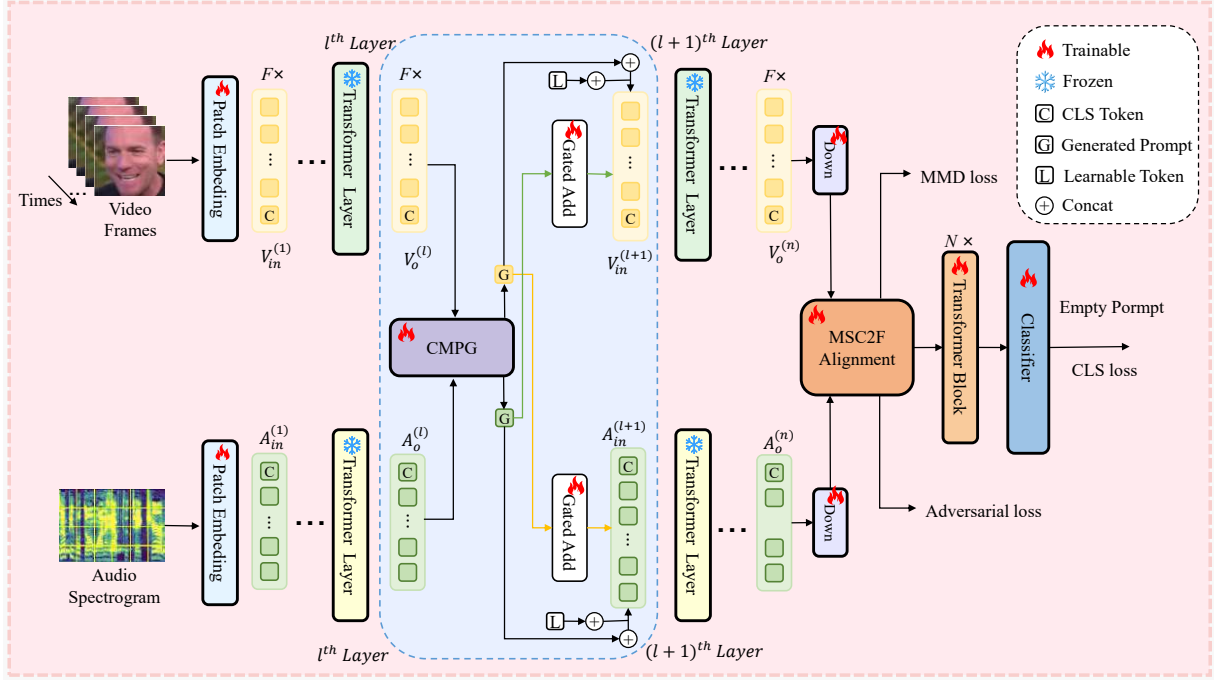


Figure 1: The architecture of BHGap. Audio and video features are extracted by frozen ViT-based backbones, while an SSM-based cross-modal prompt generator (CMPG) enables iterative co-evolution at each encoding layer. A two-stage alignment module (MSC2F) explicitly calibrates heterogeneous features via low-rank adversarial and MMD-driven fine-grained alignment, followed by lightweight fusion for emotion classification. "Down" denotes the down-sampling of the feature dimension.

sequence is first processed by the CMPG to generate guidance prompts, which are then injected into the subsequent layer.

As shown in Fig. 2, given the ℓ -th layer outputs $V^{(\ell)}$ and $A^{(\ell)}$ from visual and audio encoders, local feature enhancement is first applied through a Patch-Level ConvBlock that restores spatial locality lost by ViT flattening. The sequence is reshaped into a pseudo-2D feature map, processed with depthwise separable convolution, and projected back to the original dimension, yielding locally enhanced features $\tilde{V}^{(\ell)}$ and $\tilde{A}^{(\ell)}$. These are concatenated and fed into a Mamba State Space Model (SSM) to capture long-range temporal dependencies. The SSM output $Z^{(\ell)}$ is decomposed into modality-specific guidance prompts $p_v^{(\ell)}$ and $p_a^{(\ell)}$, which are then injected into subsequent layers via concatenation.

For each modality, the corresponding dynamic prompt is concatenated to the original feature sequence together with a learnable static token prompt $t_a^{(\ell)}$, forming an augmented sequence that is fed into the next Transformer Layer. Then, self-modal recalibration is applied with the learnable gate parameter λ_a (in the case of the auditory branch):

$$A_{in}^{(\ell+1)} = [A_o^{(\ell)}; t_a^{(\ell)}; p_a^{(\ell)}] + \lambda_a \cdot p_v^{(\ell)}, \quad (2)$$

where $[\cdot; \cdot]$ denotes concatenation.

Through the recurrent cycle of feature encoding $\rightarrow$ prompt generation $\rightarrow$ guided update, each layer refines modality representations based on reciprocal feedback. This deep iterative "co-evolution" allows the audio and visual streams to progressively

align their temporal and semantic dynamics, mitigating early fusion noise and fostering robust joint representations. Consequently, SDIC establishes a continuous, bidirectional learning process that lays the foundation for the subsequent multi-stage alignment.

Further technical details and discussion regarding SDIC can be found in **Appendix A : More Details of SDIC**.

3.5 Multi-Stage Coarse-to-Fine Feature Alignment (MSC2F)

Visual and auditory modalities are inherently heterogeneous: video information primarily encodes two-dimensional spatial features, whereas audio spectrograms emphasize temporal and frequency characteristics. As a result, even after being processed by powerful feature extractors, these modalities still reside in distinct manifold spaces, exhibiting substantial differences in their feature distributions. Consequently, semantic consistency at fine-grained levels is difficult to guarantee. Such discrepancies may prove critical when distinguishing emotions with subtle differences (e.g., frustration vs. disgust). Direct fusion under these conditions risks producing pseudo-alignment, where one modality dominates decision-making despite rich multimodal input.

To address this issue, as shown in Fig. 3, we design a multi-stage, coarse-to-fine feature alignment module, whose objective is to explicitly calibrate feature distributions across modalities before final fusion and classification. The module comprises two

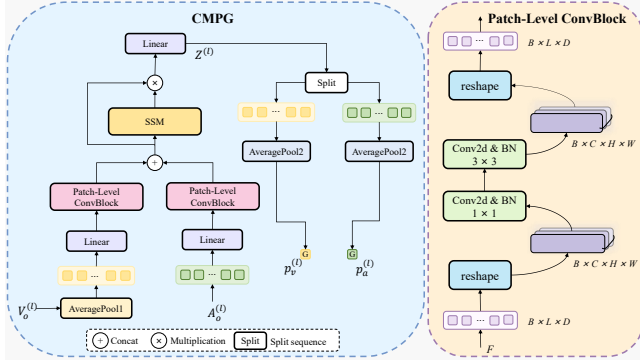


Figure 2: The architecture of the Cross-Modal Prompt Generator (CMPG). Given audio and visual features, CMPG enhances local context via Patch-Level ConvBlock (right subfigure) and performs cross-modal sequence modeling through an SSM to generate modality-specific guidance prompts. "AveragePool1" and "AveragePool2" denote the average pooling of sequence and feature dimensions, respectively.

stages: (1) **Low-rank adversarial alignment** for macro-level distribution matching and spatio-temporal congruity. (2) **Implicit differentiation-based fine-grained alignment** for micro-level statistical matching.

Stage 1: Low-Rank Adversarial Alignment. Given video and audio features ($V_o^{(n)}, A_o^{(n)}$), we first map the inputs into a shared latent space, yielding (V_D, A'), and subsequently reshape the temporal dimension of the visual feature V_D by flattening its frame sequence into V' . To address spatio-temporal heterogeneity, we apply a trainable low-rank spatio-temporal transformation parameterized by matrices $P \in \mathbb{R}^{m \times r}$ and $Q \in \mathbb{R}^{r \times n}$ ($r \ll m, n$):

$$T = PQ, \quad A'' = TA'. \quad (3)$$

This factorization establishes a compact cross-modal correspondence while suppressing redundant signals. Subsequently, we adopt a Wasserstein GAN-GP-based [16] adversarial alignment, where the generator G learns to map A'' into the video domain, and a discriminator D distinguishes $G(A'')$ from V' . This adversarial game aligns the global feature distributions while preserving modality-specific semantics, ensuring coarse spatio-temporal consistency.

Stage 2: Implicit Differentiation-based Fine-Grained Alignment. Although LRA achieves macro-level congruity, subtle discrepancies in local semantics often remain. To refine such inconsistencies, we introduce Implicit Differentiation-based Alignment (IDA), which iteratively adjusts feature representations through a latent residual correction process. Specifically, given the globally aligned features (V', A''), we define an implicit semantic coupling between modalities by introducing a residual variable Δ , which captures fine-grained semantic deviation in the latent space:

$$A''' = \hat{V} = V' + \Delta, \quad (4)$$

where Δ is not explicitly parameterized but implicitly optimized to bridge semantic inconsistencies between modalities. The update of Δ follows an implicit differentiation rule:

$$\Delta^{(t+1)} = \Delta^{(t)} - \eta \nabla_{\Delta} \mathcal{L}_{\text{align}}(V' + \Delta^{(t)}, A'''^{(t)}), \quad (5)$$

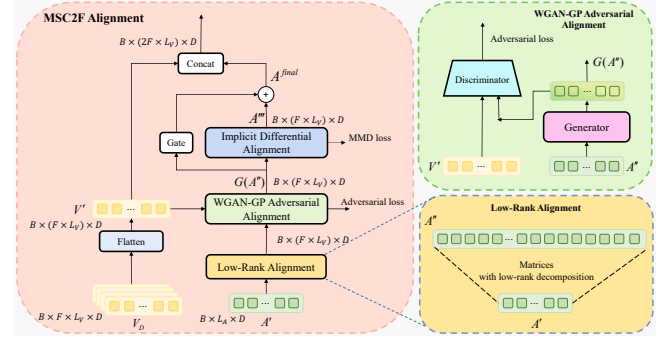


Figure 3: The multi-stage coarse-to-fine feature alignment module. Our design progressively calibrates heterogeneous audio-visual features: Stage 1 establishes global spatio-temporal congruity through low-rank adversarial training, while Stage 2 enforces fine-grained semantic consistency via MMD-driven implicit differentiation.

which iteratively corrects mismatched representations in a high-dimensional latent space. Through this process, Δ serves as a semantic mediator, enabling fine-grained information transfer between audio and visual features. To explicitly model semantic consistency during this refinement process, we further employ a lightweight semantic similarity network that projects both $A'''^{(t)}$ and V' into a shared semantic embedding space to update $A'''^{(t)}$. This network facilitates semantic comparison at the feature level and stabilizes the implicit optimization trajectory.

After convergence, we enforce statistical consistency between the refined representations by computing the Maximum Mean Discrepancy (MMD) [13] between the distributions of A''' and V' , encouraging alignment of higher-order feature moments within a reproducing kernel Hilbert space. Finally, the output of IDA is integrated with the coarse-aligned feature $G(A'')$ through a gated residual mechanism, adaptively balancing global distribution alignment and fine-grained semantic correction, yielding the final aligned audio representation A^{final} .

Finally, A^{final} and V' are concatenated and passed through multiple Transformer blocks for temporal fusion. A linear classifier is then applied for the final emotion recognition.

Further technical details and discussion regarding MSC2F can be found in **Appendix B: More Details of MSC2F**.

3.6 Loss Function

To optimize the proposed *BHGap* framework, we design a multi-component loss function that supervises the training process at different stages and ensures both modality alignment and classification performance. The overall loss is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cls}} + \lambda_{\text{adv}} \mathcal{L}_{\text{adv}} + \lambda_{\text{mmd}} \mathcal{L}_{\text{mmd}}, \quad (6)$$

where $\mathcal{L}_{\text{cls}}$ is the loss of the cross-entropy classification, $\mathcal{L}_{\text{adv}}$ is the adversarial alignment loss, and $\mathcal{L}_{\text{mmd}}$ is the fine-grained alignment loss based on the maximum mean discrepancy. λ_{adv} and λ_{mmd} are hyperparameters that control the relative importance of the corresponding terms.

Table 1: Performance comparison on the DFEW dataset. The best results are highlighted in bold, and the runner-up results are underlined. The symbol "-" indicates that the original result was not reported or could not be reproduced. The acronyms V, A, and T denote the Visual, Audio, and Text modalities, respectively.

Methods	Venues	Modalities	Accuracy of Each Emotion(%)								Metrics(%)	
			Hap.	Sad	Neu.	Ang.	Sur.	Dis.	Fea.		WAR	UAR
IAL[22]	AAAI 2023	V	87.95	67.21	70.10	76.06	62.22	0.00	36.44		69.24	55.71
M3DFEL[34]	CVPR 2023	V	89.59	68.38	67.88	74.24	59.69	0.00	31.63		69.25	56.10
MAE-DFER[31]	ACM MM 2023	V	92.92	77.46	74.56	76.94	60.99	18.62	42.35		74.43	63.41
S2D[5]	Affective Comput. 2024	V	93.87	<u>83.25</u>	75.31	<u>84.19</u>	64.33	0.00	37.07		75.98	62.57
HDF[8]	ACM MM 2025	V	89.67	71.20	67.42	73.03	64.44	12.41	41.63		71.60	60.40
ViT_LSTM+D2SP[35]	CVPR 2025	V	-	-	-	-	-	-	-		69.20	55.61
DFER-CLIP[42]	BMVC 2023	V+T	91.12	75.34	71.15	74.09	56.30	11.72	37.81		71.25	59.61
EmoCLIP[10]	FG 2024	V+T	-	-	-	-	-	-	-		62.12	58.04
DK-CLIP[24]	ACM MM 2024	V+T	<u>94.61</u>	76.19	75.06	78.65	63.61	<u>23.60</u>	<u>42.63</u>		75.41	64.95
FineCLIPER[4]	ACM MM 2024	V+T	93.95	83.92	<u>76.24</u>	84.23	<u>64.35</u>	20.47	38.72		76.21	65.98
TMEP[40]	TCSVT 2023	V+A	-	-	-	-	-	-	-		68.85	57.16
HiCMAE-S[32]	INFORM FUSION 2024	V+A	-	-	-	-	-	-	-		74.33	63.05
HiCMAE-B[32]	INFORM FUSION 2024	V+A	-	-	-	-	-	-	-		75.01	63.76
MMA-DFER[7]	CVPR 2024	V+A	-	-	-	-	-	-	-		<u>77.51</u>	<u>67.01</u>
AVF-MAE++(B)[36]	CVPR 2025	V+A	-	-	-	-	-	-	-		75.42	63.74
AVF-MAE++(L)[36]	CVPR 2025	V+A	-	-	-	-	-	-	-		76.24	65.14
VAEmo[6]	arXiv 2025	V+A	-	-	-	-	-	-	-		75.78	64.02
BHGap(ours)	-	V+A	95.46	82.47	77.51	81.59	67.45	28.96	49.57		78.80	69.03

4 Experiments

In this section, we conduct experiments to assess the performance of our model in the task of DFER. Specifically, we aim to answer the following three research questions:

- RQ1: How does the method proposed in this study perform on the main multimodal DFER datasets?
- RQ2: What influences do various core modules exert on the overall performance of the model?
- RQ3: How do the key hyperparameters in the model impact its performance?

4.1 Datasets

Following prior studies, two widely used datasets, DFEW [20] and MAFW [25], are selected for evaluation. Both are collected from real-world Web videos across different countries and contain rich environmental interference, making them highly challenging benchmarks. Their basic information is listed as follows.

- **DFEW** consists of 16,372 video-audio clips, covering 7 basic emotion categories: happiness, sadness, anger, neutral, surprise, disgust, and fear.
- **MAFW** contains 10,045 video-audio clips with annotations for 11 more fine-grained emotion categories, including anger, disgust, fear, happiness, neutral, sadness, surprise, contempt, anxiety, and helplessness.

4.2 Evaluation Metrics

We adopt Weighted Average Recall (WAR) and Unweighted Average Recall (UAR) as metrics, consistent with the dataset benchmarks. We adopted the definitions of these two metrics in accordance with

prior work [20, 25]. The detailed setup of the experiments can be found in **Appendix C: Experimental Setup**.

4.3 Results and Analysis

4.3.1 Performance Comparison (RQ1).

Results on the DFEW dataset. Table 1 reports the comparison results on DFEW. *BHGap* consistently outperforms existing approaches in both WAR and UAR. Specifically, compared with the previous best multimodal model MMA-DFER, *BHGap* achieves improvements of +1.3% WAR and +2.0% UAR. Compared with the strongest vision-text model FineCLIPER, *BHGap* gains +2.6% WAR and +3.0% UAR. Compared with visual-only methods, our approach achieves even larger margins, demonstrating the necessity of leveraging the audio modality.

It is noteworthy that some competing methods, such as VAEmo and FineCLIPER, additionally employ MLLMs for video understanding and auxiliary descriptions, while others, like FineCLIPER and S2D, rely on external visual models for landmark detection. In contrast, *BHGap* operates solely on raw audio-visual inputs yet achieves state-of-the-art performance.

We also report class-wise recognition accuracy. For low-resource categories such as *Disgust* (1.22% samples) and *Fear* (8.14% samples), where prior methods suffer from poor recognition (sometimes near-zero accuracy), *BHGap* improves accuracy by +3.2% and +6%, respectively, compared to the second-best method, demonstrating stronger learning and representational capability in imbalanced settings.

Results on the MAFW dataset. As shown in Table 2, *BHGap* again outperforms all competitors. Compared with VAEmo, the previous best multimodal model, *BHGap* improves WAR and UAR by +1% and +2%, respectively. Given that MAFW contains more

Table 2: Performance comparison on the MAFW dataset.

Methods	Modalities	Metrics(%)	
		WAR	UAR
Former-DFER[41]	V	43.27	31.16
MAE-DFER[31]	V	54.31	41.62
S2D[5]	V	57.37	43.40
ViT_LSTM+D2SP[35]	V	42.58	29.85
DFER-CLIP[42]	V+T	52.55	39.89
EmoCLIP[10]	V+T	41.46	34.24
DK-CLIP[24]	V+T	56.56	43.01
FineCLIPER[4]	V+T	56.91	45.01
T-ESFL[25]	V+A	48.18	33.35
TMEP[40]	V+A	51.15	37.17
HiCMAE-S[32]	V+A	54.45	41.66
HiCMAE-B[32]	V+A	56.17	42.65
MMA-DFER[7]	V+A	58.52	44.11
AVF-MAE++(B)[36]	V+A	57.50	43.10
AVF-MAE++(L)[36]	V+A	59.13	45.36
VAEmo[6]	V+A	58.91	45.67
BHGap(ours)	V+A	59.97	47.68

fine-grained categories (11 emotions), these results confirm the superior generalization ability of *BHGap* across diverse emotion categories.

4.3.2 Ablation Study (RQ2).

Table 3 reports the results of ablation on each module: SDIC (SSM-based deep iterative co-evolution strategy), LRA (Low-rank adversarial alignment), IDA (MMD-driven implicit differentiation alignment), and Fusion network (Transformer stack).

It can be observed that, among all modules, the fusion network exerts the most significant impact on the overall model. When the fusion module is removed, the performance drops below that of certain unimodal models. This is attributed to the fact that the final fusion network serves as the core of the multimodal framework, facilitating large-scale interaction and integration of audio-visual features. Furthermore, we observe that removing all cross-modal alignment strategies (Model G) leads to a considerable decline in performance. This underscores that heterogeneous modalities exhibit inherent feature distribution discrepancies and semantic disparities. Direct large-scale feature fusion without alignment often fails to leverage the strengths of each modality and may even introduce interference between differently structured features.

Adding either LRA or IDA to Model G improves cross-modal alignment quality to some extent, yielding noticeable performance gains. Moreover, based on the results of Models A, B, and C, it is evident that incorporating only the SDIC strategy—even in the absence of any multimodal alignment or fusion structures—still brings about an approximate 2.0+% improvement over the baseline. This can be attributed to SDIC’s cross-modal prompting and interaction mechanism, which enables a certain degree of multimodal information exchange and supports adaptive transfer from pre-training data to the target dataset, thereby enhancing suitability for the audio-visual emotion recognition task. A slight performance drop is observed when SDIC is ablated, further validating the necessity of

Table 3: Ablation study on different modules.

Models	SDIC	LRA	IDA	Fusion	WAR(%)	UAR(%)
A	×	×	×	×	72.34	59.43
B	✓	×	×	×	74.52	62.01
C	×	✓	✓	✓	78.30	68.11
D	✓	✓	×	✓	78.21	68.27
E	✓	×	✓	✓	78.03	68.15
F	✓	×	×	✓	77.57	67.12
G	✓	✓	✓	×	74.99	62.55
H	✓	✓	✓	✓	78.80	69.03

Table 4: Comparison of different prompting methods.

Methods	WAR(%)	UAR(%)
ILM-VP[3]	73.26	59.84
Mona[37]	45.78	38.36
MMA-DFER[7]	74.58	61.96
SDIC w/o CMPG	73.22	59.55
SDIC(ours)	75.40	62.53

early-stage shallow interaction in multimodal emotion recognition.

Comparison of Prompting Methods. We further investigate different prompting strategies. For fair comparison, all methods were evaluated on DFEW Fold1 under identical conditions. To better isolate the effect of prompting, all approaches employed the same backbone network with fixed pre-trained weights, excluded any cross-modal alignment structures, and fused features from different modalities solely via simple averaging. Detailed results are presented in Table 4.

Specifically, ILM-VP fine-tunes the model by adding "small patches" to the original input images; Mona is a plug-and-play fine-tuning domain adaptation adapter inserted into frozen Transformer layers for tuning; MMA-DFER employs a progressive prompting strategy combined with a Bottleneck structure for fine-tuning and cross-modal interaction; SDIC w/o CMPG ablates the cross-modal prompt generator at each layer from the current model, effectively reducing it to the classical VPT (Visual Prompt Tuning) [19] approach.

As shown in Table 4, methods enabling layer-wise cross-modal interaction consistently outperform those without information flow (ILM-VP, Mona, SDIC w/o CMPG). This confirms the importance of deep iterative cross-modal co-evolution. This underscores the importance of deep, iterative multimodal co-evolution in audiovisual Dynamic Facial Expression Recognition (DFER) tasks, which facilitates collaborative learning between the auditory and visual modalities. Although MMA-DFER incorporates layer-wise interaction via a Bottleneck structure, it simply adds large volumes of cross-modal features to the current modality’s features without adequately addressing the challenge of heterogeneous modality alignment. This likely contributes to its suboptimal performance.

Further Analysis of Alignment Modules. To more intuitively examine the effectiveness of our proposed multi-stage multimodal feature alignment strategy, we visualize the feature distributions using a representative example from the *happy* expression category,

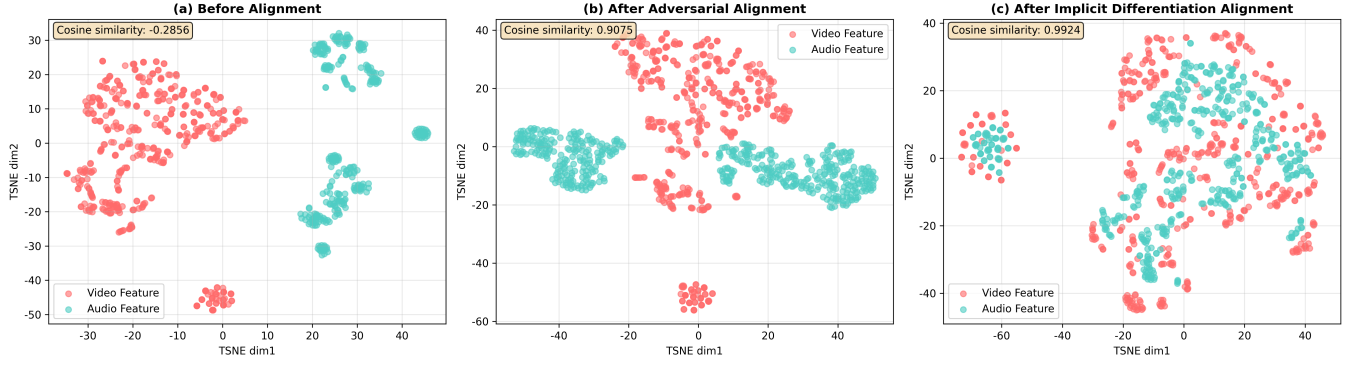


Figure 4: t-SNE visual analysis of multi-stage alignment strategies. Red represents video features, green represents audio features, and the top left corner of each figure shows the cosine similarity between the two modalities. (a) Features after encoder feature extraction and before modal alignment. (b) Features after the low-rank adversarial alignment module. (c) The aligned features. MSC2F obviously narrows the distance between modalities, reduces the difference between modalities, and increases the fine-grained semantic consistency of the two modalities.

while other emotion categories exhibit consistent trends. Specifically, we perform a t-SNE dimensionality reduction on features from different stages of the *BHGap* and measure the semantic consistency between the two modalities using cosine similarity. Fig. 4(a) shows the feature distributions of the audio and video modalities before alignment, where the two modalities are clearly separated in the shared space and even demonstrate a negative cosine correlation, indicating that early fusion may introduce mutual interference. Fig. 4(b) illustrates the distributions after Low-Rank Adversarial alignment, in which the macro-level calibration substantially reduces the modality gap and leads to highly consistent semantic representations. Fig. 4(c) presents the final aligned features, where both modalities are closely clustered within the same semantic space, reflecting strong cross-modal compatibility. Notably, although cosine similarity is not explicitly optimized as a loss function, our multi-stage alignment framework effectively calibrates semantic consistency and significantly mitigates the discrepancy between audio and video modalities.

We further examine the effects of the residual gating mechanism and Low-Rank Spatio-Temporal Transformation (LowRank-STT) in the MSC2F module. As shown in Table 5, removing the residual gate results in purely sequential alignment, where audio features overfit video features under MMD-driven implicit alignment, losing modality-specific characteristics, particularly distinctive emotional cues. Retaining the residual connection propagates coarse-aligned audio features and preserves modality uniqueness. We also observe performance drops in WAR and UAR when LowRank-STT is ablated, as this component aligns audio sequence length with video features that merge spatial and temporal dimensions. Without it, audio sequences become much shorter, compressing temporal information and impairing the ability to capture subtle emotional dynamics, ultimately degrading recognition accuracy.

4.3.3 Hyperparameter Test (RQ3).

In our experiments concerning hyperparameters, we investigated the effects of the following aspects on the results: (1) the method of multimodal aggregation, (2) the number of randomly sampled

Table 5: Ablation of low-rank spatio-temporal transformation and gating strategy in MSC2F.

Methods	WAR(%)	UAR(%)
MSC2F w/o gate	76.41	66.20
MSC2F w/o LowRank-STT	75.68	64.57
MSC2F	78.80	69.03

points used for approximating the MMD loss, (3) the dimensionality of the hidden space for multimodal alignment and fusion, (4) the rank of the low-rank matrices in the low-rank spatiotemporal transformation, and (5) the number of layers in the stacked Transformer blocks for multimodal fusion. Through comprehensive hyperparameter experiments, we identify the optimal model and experimental configuration. Detailed experimental results and analysis are provided in **Appendix D: Hyperparameter Analysis**.

5 Conclusion

Dynamic facial expression recognition (DFER) plays an important role in media content mining and human-computer interaction. In this work, we propose *BHGap*, a novel framework that addresses key challenges in DFER by emphasizing shallow cross-modal interactions and integrating deep iterative prompt generation with a multi-stage coarse-to-fine alignment strategy. An SSM-based cross-modal prompt generator enables dynamic and fine-grained cooperative learning between audio-visual features, while the alignment module explicitly bridges the modality gap at both macro and micro levels. Extensive experiments on the DFEW and MAFW datasets demonstrate the effectiveness of our framework and its state-of-the-art performance. Future work will explore extending this paradigm to other multimodal affective computing tasks and broader multimodal scenarios.

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Appendix A : More Details of SDIC

Conventional audio-visual fusion models usually perform one-shot fusion (e.g., concatenation or top-layer cross-attention), neglecting deep-layer interactions. To emulate the natural co-evolution of human perceptual cues, our framework establishes a fine-grained deep iterative co-evolution mechanism, in which each layer of the frozen encoders exchanges information via a Cross-Modal Prompt Generator (CMPG) based on State Space Models (SSM).

A.1 Design Motivation

Since the Vision Transformer (ViT) flattens visual and audio spectrogram inputs into 1D patch sequences, it loses spatial locality. To restore local inductive bias before cross-modal dependency modeling, CMPG includes a Patch-Level ConvBlock, which reshapes the sequence into a pseudo-2D map, applies depthwise separable convolution (1×1 pointwise + 3×3 depthwise), and projects back to the original dimension. This produces locally enhanced features $\tilde{V}^{(\ell)}$ and $\tilde{A}^{(\ell)}$ for each layer.

A.2 State Space Modeling within CMPG

The enhanced features from both modalities are concatenated and fed into an SSM for long-range dependency modeling. The continuous form of SSM is:

$$\dot{h}(t) = Ah(t) + Bu(t), \quad y(t) = Ch(t) + Du(t), \quad (7)$$

where $u(t)$ denotes input sequence, $y(t)$ is the output, and A represents the state transition matrix.

Through zero-order hold discretization, this model can be transformed into a discrete parallel form. We employ the selective scan mechanism of Mamba, where the transition matrix A and step size Δ are dynamically generated based on the input, enabling the model to selectively attend to informative temporal segments while ignoring redundant ones. This ensures efficient sequence modeling and captures long-term dependencies critical for emotional co-evolution.

A.3 Role of CMPG in Iterative Co-evolution

The SSM output $Z^{(\ell)}$ is decomposed into modality-specific guidance prompts $p_v^{(\ell)}$ and $p_a^{(\ell)}$, which are injected into the next encoder layer to guide subsequent representations. Through the cyclic process of feature encoding $\rightarrow$ prompt generation $\rightarrow$ guided update, audio-visual features engage in continuous, bidirectional adaptation. This mechanism transforms fusion into a layer-wise dynamic dialogue, mitigating early-fusion noise and laying the foundation for the multi-stage alignment in Section 3.5.

Appendix B: More Details of MSC2F

This appendix provides additional clarification of the two-stage alignment process described in Section 3.5. The proposed Multi-Stage Coarse-to-Fine Feature Alignment (MSC2F) aims to bridge the inherent heterogeneity between visual and auditory representations through global-to-local alignment. Specifically, it consists of Stage 1: Low-Rank Adversarial Alignment (LRA) and Stage 2: Implicit Differentiation-based Alignment (IDA).

B.1: Low-Rank Adversarial Alignment for Spatio-Temporal Congruity

This stage aims to address two core challenges at the macro level: (1) aligning the overall feature distributions of audio and video sequences, and (2) establishing spatio-temporal congruity between

modalities. We achieve this by combining low-rank transformations with adversarial training.

Low-rank Spatio-Temporal Transformation. To construct flexible correspondences between audio and video sequences of varying lengths, we abandon rigid one-to-one alignment and instead design a soft transformation mechanism based on low-rank decomposition. The underlying assumption is that although the temporal mapping between audio and video can be complex, their intrinsic correlations can be effectively approximated within a low-dimensional subspace, which also improves robustness. This hypothesis is grounded in extensive research indicating that correlations among temporal signals are often concentrated in a few dominant directions, which can be effectively approximated by a low-rank subspace [12]. Such mapping not only suppresses noise and redundant features but also enhances the generalization capability under few-shot learning conditions.

Formally, given video features $V_o^{(n)}$ and audio features $V_o^{(n)}$, we first project both modalities into a shared lower-dimensional space and flatten the temporal dimension of video features, yielding spatio-temporally adjusted video features V' and audio features A' . Next, we introduce two learnable low-rank basis matrices $P \in \mathbb{R}^{m \times r}$ and $Q \in \mathbb{R}^{r \times n}$, where r is a predefined rank much smaller than the sequence lengths m and n . By dynamically slicing these matrices according to actual sequence lengths in each batch, we construct a transformation matrix:

$$T = PQ, \quad (8)$$

where each element T_{ij} represents the importance of the i -th audio timestep to the j -th video frame. Using T , we linearly resample and align audio features to the video sequence:

$$A'' = TA'. \quad (9)$$

This low-rank parameterization not only reduces the number of learnable parameters but also reformulates the temporal alignment task as learning a low-dimensional latent correlation space, thereby enabling explicit and efficient modeling of spatio-temporal congruity.

Adversarial Distribution Alignment. Although the above transformation aligns sequence lengths, the distributions of A'' and V' still differ significantly. A naive solution, such as mean squared error, would force direct point-wise alignment, but this tends to yield blurred, averaged solutions and ignores one-to-many mappings in real emotional expressions (e.g., the same sad expression may co-occur with either sobbing or humming). To overcome this limitation, we adopt a Generative Adversarial Network (GAN) paradigm.

Specifically, we employ the WGAN-GP framework, which minimizes the Wasserstein distance between distributions, thereby ensuring stable training and mitigating mode collapse—problems that often arise in vanilla GANs. Compared with other GAN variants, WGAN-GP with gradient penalty guarantees the 1-Lipschitz constraint of the discriminator, making it particularly suitable for tasks with substantial cross-modal distribution gaps.

In our implementation, the generator G (a lightweight multilayer perceptron) maps audio features A'' into an aligned space, while the discriminator D (a deep network with linear layers, LeakyReLU, and Dropout) distinguishes real video features V' from generated features $G(A'')$. To further encourage semantic consistency, we

introduce a feature matching loss defined as mean squared error between $G(A'')$ and V' . The overall adversarial loss is:

$$\mathcal{L}_{adv} = \mathcal{L}_{WGAN} + \mathcal{L}_{GP} + \alpha \mathcal{L}_{FM}, \quad (10)$$

where $\mathcal{L}_{WGAN}$ is the Wasserstein loss, $\mathcal{L}_{GP}$ is the gradient penalty term, and $\mathcal{L}_{FM}$ is the feature matching loss. Here α is set to a small value, i.e. 0.1, to ensure the model focuses primarily on cross-modal alignment.

B.2: Implicit Differentiation for Fine-grained Alignment

Although adversarial training in Stage 1 can align the macro-level distributions between video and audio, it depends heavily on the feedback of the discriminator and thus lacks sensitivity to high-order statistical moments. As a result, subtle semantic misalignments between modalities may persist. This is especially problematic in fine-grained emotion recognition, where distinguishing between categories such as fear and surprise requires alignment at a more refined level than global distribution matching. To address this limitation, we design a fine-grained alignment stage based on implicit differentiation, assisted by a semantic consistency network.

After Stage 1, the adversarially aligned audio features are denoted as $G(A'')$. However, $G(A'')$ still contains tokens of varying semantic quality—some tokens are strongly correlated with video features, while others represent noise or irrelevant components. To refine these features, we employ a semantic consistency network (a lightweight MLP) that computes token-level similarity weights between video and audio representations. Concretely, at each iteration, given the current refined video representation $V' + \Delta$, the network outputs similarity weights via a sigmoid function. These weights adaptively highlight semantically important tokens while suppressing less informative ones. Multiplying the raw audio tokens with these weights yields the re-weighted audio features A''' . Importantly, A''' is dynamically updated at each iteration, rather than fixed after a single step.

Instead of directly aligning V' with A''' , we introduce a residual variable Δ to refine the video features:

$$A''' = \hat{V} = V' + \Delta, \quad (11)$$

The residual Δ is solved iteratively through an implicit differentiation process. At each iteration t , $A'''^{(t)}$ is recomputed from the semantic consistency network, and Δ is updated according to:

$$\Delta^{(t+1)} = \Delta^{(t)} - \eta \cdot \nabla f(V' + \Delta^{(t)}, A'''^{(t)}), \quad (12)$$

where η is the learning rate, and $f(\cdot)$ is the alignment objective defined by the similarity network. Through this iterative process, video features are gradually adapted to the semantically consistent audio representation. After convergence, we obtain the refined video features $\hat{V}$, which can be regarded as the implicitly aligned representation. Correspondingly, the dynamically updated A''' serves as the fine-grained aligned audio representation, matched to $\hat{V}$.

To further ensure statistical consistency, the Maximum Mean Discrepancy (MMD) is adopted as the distance metric between the video features V' and the audio features A''' after implicit differentiation alignment. MMD is a kernel-based integral probability metric capable of aligning all statistical moments of two distributions. Given two sets of features X, Y sampled from the distribution, the

unbiased empirical estimate is as follows.

$$\begin{aligned} L_{MMD}(X, Y) = & \frac{1}{m(m-1)} \sum_{i \neq j} k(\mathbf{x}_i, \mathbf{x}_j) + \frac{1}{n(n-1)} \sum_{i \neq j} k(\mathbf{y}_i, \mathbf{y}_j) \\ & - \frac{2}{mn} \sum_{i,j} k(\mathbf{x}_i, \mathbf{y}_j), \end{aligned} \quad (13)$$

where $k(\cdot, \cdot)$ denotes a Gaussian kernel. In practice, we adopt a multi-scale Gaussian kernel to enhance robustness and capture discrepancies across different scales. The corresponding loss is defined as:

$$\mathcal{L}_{mmd} = \text{MMD}^2(V', A'''). \quad (14)$$

B.3 Residual Gating Fusion

To balance the contributions from coarse adversarial alignment $G(A'')$ and fine-grained implicit refinement A''' , we design a gated residual mechanism:

$$A^{final} = \gamma G(A'') + (1 - \gamma) A''', \quad (15)$$

where γ is a learnable gating parameter. This mechanism ensures that the final audio representation integrates both stages: the global distributional consistency enforced by adversarial training, and the fine-grained semantic refinement achieved via implicit differentiation.

Appendix C: Experimental Setup

All experiments are implemented in PyTorch and conducted on a single NVIDIA A100-40G GPU.

For video input, we extract 16 consecutive frames of size 224×224 pixels from each video clip. Audio signals are converted into Mel-spectrograms before being fed into the audio encoder. For MAFW, we follow the protocol in MMA-DFER [7], applying MTCNN to extract face frames.

During training, we fix the random seed to 1, use a batch size of 8, and set the initial learning rate to $3e-5$. The AdamW optimizer with weight decay $1e-2$ is employed, and the learning rate follows a cosine decay schedule. The backbone encoders remain fully frozen throughout the training. The total number of epochs is 25.

Regarding the loss function, we emphasize the centrality of the classification task while balancing alignment objectives. After searching $\lambda_{adv} \in [0.01, 0.7]$ and $\lambda_{mmd} \in [0.01, 0.7]$, we set $\lambda_{adv} = 0.2$ and $\lambda_{mmd} = 0.1$ for the final configuration.

The multimodal fusion network is implemented with three Transformer layers. Before alignment, audio and video features are projected into a 512-dimensional space. The rank of the low-rank matrices in the spatio-temporal transformation is set to 8. For implicit differentiation alignment, the maximum number of iterations is set to 3. For MMD loss computation, 1024 tokens are randomly sampled per step.

Both datasets are evaluated under 5-fold cross-validation to ensure fairness.

Appendix D: Hyperparameter Analysis

Comparison of Fusion Methods. In Table 6, we compare the effects of different modality fusion methods. Add and Concat are simple fusion approaches, which directly sum or concatenate the features of different modalities, respectively. Linear, Mamba, and Transformer in this experiment all employ a 3-layer structure to

Table 6: Comparison of multimodal feature aggregation using different methods. The Transformer architecture yields the best results.

Aggregation Methods	WAR(%)	UAR(%)	Params(M)
Add	74.99	62.55	0.00
Concat	74.49	62.09	0.00
Linear	76.52	65.74	6.30
Mamba	78.25	68.34	5.10
Transformer	78.80	69.03	9.46

Table 7: Comparison of different latent space dimensions during multimodal alignment and fusion. The best performance is achieved with hidden_dim=512.

hidden_dim	WAR(%)	UAR(%)	Params(M)
64	75.96	64.55	0.25
128	77.32	64.55	0.81
256	78.25	67.76	2.83
512	78.80	69.03	10.51
768	78.55	67.20	23.04

fuse the concatenated features from different modalities. It can be observed that the simple Add and Concat methods, which lack sophisticated multi-modal information modeling capabilities, perform relatively poorly. The fusion method using a linear layer is inferior to the attention-based Mamba and Transformer architectures. In the complex task of audiovisual dynamic facial expression recognition, Transformer’s attention modeling for long sequences is marginally superior to that of Mamba. However, the parameter count of Mamba is reduced by 46% compared to a Transformer with the same number of layers, and is even lower than that of the linear layer fusion method. Considering the acceptable parameter cost, we select the Transformer as the final fusion structure and designate the Mamba fusion as a faster version of the model.

Effect of Hidden Dimension. A comparative study on the dimensionality of the hidden layer used for multimodal alignment and fusion was conducted, with the results presented in Table 7. Optimal hidden dimension for multimodal fusion is 512. An excessively low-dimensional space, while significantly reducing the number of parameters, may incur an unacceptable loss of feature information. Conversely, an excessively high dimensionality renders the representations of different modalities in the shared space more complex. This increased complexity of modal information can compromise the model’s ability to effectively discover inter-modal relationships, thereby imposing limitations on the performance of multimodal alignment and fusion.

Effect of MMD Sampling. To reduce the computational overhead associated with calculating the MMD loss, we employ a random sampling strategy. In Table 8, we compare the impact of the number of samples per training step on the final performance. It can be observed that when the number of samples per step reaches 256 or more, the final results are nearly identical. This indicates that the MMD loss computed using a sufficient number of random samples can effectively approximate the loss calculated on the entire batch.

Table 8: Comparison of the effect of different sample sizes on the results when calculating MMD loss. The best performance is achieved with sample_size=1024.

sampling size	WAR(%)	UAR(%)
128	78.22	68.13
256	78.47	68.53
512	78.62	68.91
1,024	78.80	69.03
2,048	78.59	68.71

Table 9: Comparison of the influence of the rank of the low-rank matrix on the results in low-rank spatio-temporal transformation. The best performance is achieved with rank=8.

rank	WAR(%)	UAR(%)
4	78.27	68.64
8	78.80	69.03
16	78.34	68.73
32	78.50	68.79
64	78.59	68.82

Table 10: Compare the effect of the number of Transformer layers on the results. Optimal performance is achieved when the network is configured with 3 layers.

Number of layers	WAR(%)	UAR(%)
1	78.44	68.33
2	78.52	68.73
3	78.80	69.03
4	78.31	68.92
5	78.34	68.69

Effect of the Rank of Low-Rank Matrix. In Table 9, we compare the impact of employing different ranks for the low-rank matrix in the Low-Rank Spatio-Temporal Transformation on the experimental results. We find that the rank of the low-rank matrix has less influence on the overall result. Optimal performance across all metrics is achieved when the rank of the matrix is set to 8. The low-rank matrix compels the model to perform the cross-modal temporal alignment task in a highly compressed manner; however, an excessively low rank may limit the model’s capacity to fit the underlying feature transformations. Conversely, setting the rank higher allows the model to learn more complex alignment relationships, but this increases susceptibility to overfitting.

Effect of the Number of Transformer Layers. Table 10 presents the results of employing Transformer Blocks with varying numbers of layers for feature fusion. Our analysis indicates that performance reaches its optimum with the layer count set to 3. An inadequate number of layers results in deficient fusion modeling, while an excessive count introduces superfluous parameters, complicating optimization and demanding more intricate tuning. This robustness is not only a consequence of the Transformer’s formidable sequence modeling capabilities but also serves as a testament to the efficacy of the proposed architectural framework.